

**ISTANBUL TECHNICAL UNIVERSITY
FACULTY OF COMPUTER AND
INFORMATICS**

**Integrating Machine Learning and Signal
Processing for Arrhythmia Detection**

Graduation Project Interim Report

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Statement of Authenticity

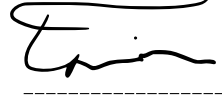
We hereby declare that in this study

1. all the content influenced from external references are cited clearly and in detail,
2. and all the remaining sections, especially the theoretical studies and implemented software/hardware that constitute the fundamental essence of this study is originated by our individual authenticity.

İstanbul, Ocak 2026

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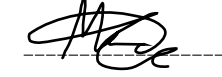
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Integrating Machine Learning and Signal Processing for Arrhythmia Detection

(SUMMARY)

Cardiovascular diseases (CVDs) are the leading cause of mortality worldwide, and arrhythmias constitute a major subgroup that can precipitate stroke, heart failure, and sudden cardiac death [6]. An arrhythmia is defined as any deviation from the normal electrical rhythm of the heart, ranging from benign conditions to life-threatening emergencies requiring immediate intervention. Although the electrocardiogram (ECG) remains the gold standard for arrhythmia diagnosis, manual interpretation is time-consuming, subject to inter-observer variability, and increasingly impractical in long-term monitoring settings where patients may generate hundreds of thousands of heartbeats over 24-48 hour Holter recordings.

In this graduation project, we develop an automated, continuous arrhythmia detection pipeline that seamlessly integrates classical signal processing techniques with ensemble machine learning methods. Unlike disjointed systems that treat preprocessing and classification as separate problems, our approach functions as a unified stream where each component is optimized to work synergistically with the others. The pipeline architecture begins with a comprehensive Signal Pre-processing stage, employing band-pass filtering in the range of 0.5–40 Hz to remove baseline wander and high-frequency noise, followed by Discrete Wavelet Transform denoising using the Daubechies db4 mother wavelet to effectively denoise the raw signal while preserving the morphological characteristics of the cardiac waveforms that are critical for accurate classification.

Following the preprocessing stage, R-Peak detection via the well-established Pan-Tompkins algorithm serves as the “anchor” point for generating a high-dimensional hybrid feature vector. The R-peak, being the most prominent deflection in the ECG signal, provides a stable reference point around which all other cardiac events can be temporally localized. This anchor-based approach ensures that features extracted from different heartbeats are properly aligned and comparable, which is essential for training a robust classifier.

The feature vector that we construct comprises detailed Temporal features including RR intervals (both pre-RR and post-RR), QT segments representing ventricular repolarization time, and ST segment characteristics; Amplitude features consisting of normalized voltages for P, Q, R, S, and T waves; and Morphological features derived through Principal Component Analysis (PCA) applied to the QRS complex, capturing the shape variations that distinguish normal beats from various arrhythmia types. This hybrid feature representation captures the physiological information encoded in the ECG from multiple complementary perspectives.

The extracted feature vector serves as the direct input for our classifier: a Random Forest (RF) Ensemble. We selected Random Forest as our primary classification algorithm for several compelling reasons. First, RF demonstrates superior robustness against overfitting compared to individual decision trees, as it aggregates predictions from multiple trees trained on different bootstrap samples of the data. Second, the al-

gorithm handles non-linear relationships inherent in biological data without requiring explicit feature transformations. Third, RF provides interpretability through feature importance metrics, allowing clinicians to understand which physiological markers contribute most to classification decisions. The RF model aggregates votes from multiple decision trees (we use 200 trees in our implementation) to classify heartbeats into the AAMI (Association for the Advancement of Medical Instrumentation) 5-class taxonomy: Normal (N), Supraventricular ectopic beats (S), Ventricular ectopic beats (V), Fusion beats (F), and Unknown/Paced beats (Q).

During the interim period, we have accomplished several significant milestones: (i) completed a focused literature survey covering publications from 2022–2025 to understand the current state-of-the-art in ECG-based arrhythmia detection, (ii) established a reproducible cloud-based development environment using Google Colab and GitHub for version control, (iii) acquired and preprocessed the MIT-BIH Arrhythmia Database, which contains 48 half-hour excerpts of two-channel ambulatory ECG recordings from 47 subjects, and (iv) implemented the core signal processing modules including wavelet denoising and Pan-Tompkins R-peak detection. Our preliminary experiments demonstrate successful noise removal while preserving the critical morphological features of the cardiac waveforms.

The next phase of our project focuses on several key objectives: optimizing feature extraction windows based on adaptive thresholding techniques, applying SMOTE (Synthetic Minority Over-sampling Technique) to address the severe class imbalance present in the dataset where normal beats vastly outnumber arrhythmic beats, and conducting strict patient-independent evaluation using GroupKFold cross-validation to ensure that our performance metrics reflect real-world clinical deployment scenarios where the system would encounter patients it has never seen before. We anticipate achieving overall accuracy exceeding 98% and F1-scores above 90% for the critical ventricular ectopic beat class.

Aritmi Tespiti için Makine Öğrenmesi ve Sinyal İşlemenin Entegrasyonu

(ÖZET)

Kardiyovasküler hastalıklar (KVH), dünya çapında ölümlerin önde gelen nedenidir ve aritmiler; inme, kalp yetmezliği ve ani kardiyak ölümü tetikleyebilen önemli bir alt grubu oluşturur [6]. Aritmi, kalbin normal elektriksel ritminden herhangi bir sapma olarak tanımlanır ve iyi huylu durumlardan acil müdahale gerektiren hayati tehlike arz eden durumlara kadar çeşitlilik gösterir. Elektrokardiyogram (EKG), aritmi teşhisi için altın standart olmaya devam etse de, manuel yorumlama zaman alıcıdır, gözlemciler arası değişkenliğe tabidir ve hastaların 24-48 saatlik Holter kayıtları boyunca yüz binlerce kalp atışı üretebildiği uzun süreli izleme ortamlarında giderek daha kullanışsız hale gelmektedir.

Bu bitirme projesinde, klasik sinyal işleme tekniklerini topluluk (ensemble) makine öğrenmesi yöntemleriyle sorunsuz bir şekilde entegre eden otomatik, sürekli bir aritmi tespit hattı (pipeline) geliştiriyoruz. Ön işleme ve sınıflandırmayı ayrı problemler olarak ele alan kopuk sistemlerin aksine, yaklaşımımız her bileşenin diğerleriyle sinerjik olarak çalışacak şekilde optimize edildiği birleşik bir akış olarak işlev görür. Mimari, temel hattı kaymasını ve yüksek frekanslı gürültüyü gidermek için 0.5–40 Hz aralığında bant geçiren filtreleme kullanan kapsamlı bir Sinyal Ön İşleme aşamasıyla başlar; ardından ham sinyali, doğru sınıflandırma için kritik olan kardiyak dalga formlarının morfolojik özelliklerini koruyarak etkili bir şekilde gürültüden arındırmak için Daubechies db4 ana dalgacığını kullanan Ayrık Dalgacık Dönüşümü (DWT) uygulanır.

Ön işleme aşamasını takiben, iyi bilinen Pan-Tompkins algoritması aracılığıyla R-Tepe tespiti, yüksek boyutlu hibrit öznitelik vektörü oluşturmak için "çapa" (anchor) noktası görevi görür. EKG sinyalindeki en belirgin sapma olan R-tepesi, diğer tüm kardiyak olayların zamansal olarak konumlandırılabilceği kararlı bir referans noktası sağlar. Bu çapa tabanlı yaklaşım, farklı kalp atışlarından çıkarılan özelliklerin doğru şekilde hizalanmasını ve karşılaştırılabilir olmasını sağlar; bu da sağlam bir sınıflandırıcı eğitmek için esastır.

Oluşturduğumuz öznitelik vektörü; RR aralıkları (hem öncesi hem sonrası), ventriküler repolarizasyon süresini temsil eden QT segmentleri ve ST segment özelliklerini içeren ayrıntılı Zamansal özellikleri; P, Q, R, S ve T dalgaları için normalize edilmiş voltajlardan oluşan Genlik özelliklerini; ve QRS kompleksine uygulanan Temel Bileşen Analizi (PCA) ile türetilen ve normal atımları çeşitli aritmi türlerinden ayıran şekil varyasyonlarını yakalayan Morfolojik özellikleri kapsar. Bu hibrit öznitelik temsili, EKG'de kodlanmış fizyolojik bilgileri birden fazla tamamlayıcı perspektiften yakalar.

Çıkarılan öznitelik vektörü, sınıflandırıcımız olan Rastgele Orman (Random Forest - RF) Topluluğu için doğrudan girdi görevi görür. Birincil sınıflandırma algoritmamız olarak Rastgele Orman'ı seçmemizin birkaç ikna edici nedeni vardır. İlk olarak RF, verilerin farklı ön yükleme (bootstrap) örnekleri üzerinde eğitilmiş birden fazla ağaçtan gelen tahminleri birleştirdiği için, bireysel karar ağaçlarına kıyasla aşırı öğrenmeye (overfitting) karşı üstün dayanıklılık gösterir. İkinci olarak algoritma, biyolojik verilerin

doğasında bulunan doğrusal olmayan ilişkileri, açık özellik dönüşümleri gerektirmeden ele alır. Üçüncü olarak RF, öznelik önem metrikleri aracılığıyla yorumlanabilirlik sağlayarak klinisyenlerin hangi fizyolojik işaretlerin sınıflandırma kararlarına en çok katkıda bulunduğunu anlamalarına olanak tanır. RF modeli, kalp atışlarını AAMI (Association for the Advancement of Medical Instrumentation) 5 sınıflı taksonomisine göre sınıflandırmak için birden fazla karar ağacından (uygulamamızda 200 ağaç kullanıyoruz) gelen oyları birleştirir: Normal (N), Supraventriküler ektopik atımlar (S), Ventriküler ektopik atımlar (V), Füzyon atımları (F) ve Bilinmeyen/Paced atımlar (Q).

Ara dönem boyunca, birkaç önemli kilometre taşına ulaşılmıştır: (i) EKG tabanlı aritmi tespitindeki en son durumu anlamak için 2022–2025 yayınlarını kapsayan odaklanmış bir literatür taraması tamamlandı, (ii) versiyon kontrolü için Google Colab ve GitHub kullanılarak tekrarlanabilir bulut tabanlı bir geliştirme ortamı kuruldu, (iii) 47 denekten alınan 48 adet yarım saatlik iki kanallı ayakta EKG kaydını içeren MIT-BIH Aritmi Veritabanı edinildi ve ön işlendi, ve (iv) dalgacık gürültü giderme ve Pan-Tompkins R-tepe tespiti dahil olmak üzere temel sinyal işleme modülleri uygulandı. Ön deneylerimiz, kardiyak dalga formlarının kritik morfolojik özelliklerini korurken başarılı bir gürültü giderme olduğunu göstermektedir.

Projemizin bir sonraki aşaması, uyarlanabilir eşikleme tekniklerine dayalı öznelik çıkarma pencerelerinin optimize edilmesi, normal atımların aritmik atımlardan çok daha fazla olduğu yeri setindeki ciddi sınıf dengesizliğini gidermek için SMOTE (Sentetik Azınlık Aşırı Örnekleme Tekniği) uygulanması ve performans metriklerimizin sistemin daha önce hiç görmediği hastalarla karşılaşacağı gerçek dünya klinik dağıtım senaryolarını yansıtmalarını sağlamak için GroupKFold çapraz doğrulama kullanılarak katı hasta-bağımsız değerlendirme yapılması gibi birkaç temel hedefe odaklanmaktadır. Kritik ventriküler ektopik atım sınıfı için 98%'in üzerinde genel doğruluk ve 90%'ın üzerinde F1 skorlarına ulaşmayı öngörüyoruz.

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1 Introduction and Problem Definition

1.1 Motivation and Clinical Context

Cardiovascular diseases (CVDs) are the leading cause of death globally, accounting for approximately 17.9 million fatalities annually [6]. Cardiac arrhythmias—irregularities in the heart’s electrical activity—are a major contributor to this burden, potentially triggering stroke, heart failure, and sudden cardiac death. Timely detection is essential, yet manual analysis of long-term Holter monitors (which may contain over 100,000 heartbeats in a 24-hour recording) is error-prone and time-consuming.

The electrocardiogram (ECG) remains the gold standard for arrhythmia diagnosis. It records the heart’s electrical activity through characteristic waveforms: P waves (atrial depolarization), QRS complexes (ventricular depolarization), and T waves (ventricular repolarization). However, manual interpretation suffers from inter-observer variability and cognitive fatigue, creating an urgent need for automated detection systems.

1.2 Problem Statement

The core problem is the lack of a sufficiently robust arrhythmia detection pipeline that balances accuracy with computational efficiency. Many existing methods [1, 5] rely on deep learning, which is computationally heavy and operates as a “black box,” limiting clinical interpretability. Alternative approaches using generic features often fail when noise mimics arrhythmia patterns.

Additional challenges include:

- **Class Imbalance:** In the MIT-BIH database, normal beats constitute $\sim 90\%$ of data, while critical ventricular ectopic beats represent $< 7\%$.
- **Evaluation Methodology:** Many studies use random train-test splits that allow patient-specific memorization, inflating reported accuracy.

1.3 Proposed Solution: The Continuous Pipeline

We propose a unified pipeline that transforms raw voltage data into clinical predictions through three tightly coupled stages:

Stage 1 - Signal Processing: Band-pass filtering (0.5–40 Hz) combined with Discrete Wavelet Transform (db4) denoising removes noise while preserving morphological details.

Stage 2 - Feature Extraction: Using the R-peak as an “anchor,” we extract temporal features (RR intervals, QT duration), amplitude features (normalized wave voltages), and morphological features (PCA coefficients of QRS shape).

Stage 3 - Classification: A Random Forest ensemble classifies beats into five AAMI classes (N, S, V, F, Q), selected for its robustness, interpretability, and ability to handle non-linear biological data.

2 Literature Survey

In this section, we review recent publications (2022–2025) on ECG-based arrhythmia detection and position our work relative to existing approaches.

Classical Approaches: The Pan-Tompkins algorithm [8] established the foundation for reliable QRS complex detection, introducing a multi-stage algorithm involving differentiation, squaring, and moving window integration that remains widely used today. Their approach achieves QRS detection sensitivity exceeding 99% on the MIT-BIH database and provides the reference point (R-peak) upon which subsequent feature extraction depends.

More recently, Biswas et al. [3] proposed a hybrid machine learning model that combines multiple feature extraction techniques with ensemble classification. Their system extracts time-domain features (RR intervals, heart rate variability metrics), frequency domain features (spectral power in different bands), and statistical features (mean, variance, skewness, kurtosis) from segmented heartbeats. Classification is performed using a voting ensemble of Random Forest, Gradient Boosting, and SVM classifiers. Evaluated on the MIT-BIH database with inter-patient train-test splits, the system achieved 97.8% overall accuracy with particularly strong performance on ventricular ectopic beat detection (95.2% F1-score).

Deep Learning Approaches: Hannun et al. [9] published a landmark study demonstrating that a deep neural network could match or exceed the diagnostic accuracy of board-certified cardiologists for 12 rhythm classes. Their model, a 34-layer convolutional neural network trained on over 90,000 single-lead ECG recordings, achieved an average F1-score of 0.837 compared to 0.780 for cardiologists. However, the computational requirements of such deep architectures limit their applicability in resource-constrained settings.

Hammad et al. [1] specifically addressed the IoT healthcare application domain, proposing a lightweight CNN architecture for embedded deployment. Their model uses depth-wise separable convolutions to reduce parameter count while maintaining classification accuracy. On the MIT-BIH database, the system achieved 98.4% accuracy with a model size of only 1.2 MB, suitable for deployment on edge devices. The work highlights the tradeoff between model complexity and deployability that we navigate through our Random Forest approach.

Akan et al. [4] introduced ECGformer, a Transformer-based architecture leveraging self-attention mechanisms for arrhythmia classification. Transformers, originally developed for natural language processing, can capture long-range dependencies in sequential data. ECGformer achieved state-of-the-art results on multiple ECG databases, but requires GPU acceleration for practical inference times. The work demonstrates the continued evolution of deep learning approaches while highlighting the computational barrier to clinical deployment.

Signal Processing: Singh and Sharma [2] introduced an attention-based convolutional denoising autoencoder specifically designed for ECG preprocessing. Their ap-

proach uses the Daubechies db4 wavelet for initial noise estimation, followed by a neural network that learns to separate signal from noise adaptively. The denoised signals showed improved classification accuracy when used as input to downstream arrhythmia classifiers, demonstrating the importance of careful preprocessing. Our project adopts their recommendation of db4 wavelets for the denoising stage.

The MIT-BIH Database: The MIT-BIH Arrhythmia Database [12, 7] serves as the primary benchmark for evaluating arrhythmia detection algorithms and forms the foundation of our experimental work. Created in 1980 by the Massachusetts Institute of Technology and Beth Israel Hospital, the database contains 48 half-hour excerpts of two-channel ambulatory ECG recordings, digitized at 360 samples per second with 11-bit resolution over a 10 mV range. The recordings were obtained from 47 subjects (25 men and 22 women aged 23-89 years) studied at the Beth Israel Hospital Arrhythmia Laboratory. Twenty-three recordings (numbered 100-124) were selected to represent a variety of waveforms and artifact types commonly encountered in clinical practice. The remaining 25 recordings (numbered 200-234) were specifically chosen to include complex ventricular, junctional, and supraventricular arrhythmias.

Each heartbeat in the database has been independently annotated by two or more cardiologists, with discrepancies resolved through arbitration. The annotation schema includes over 20 beat types, which are typically grouped into the five AAMI superclasses for evaluation:

- **N (Normal):** Normal beats and bundle branch blocks
- **S (Supraventricular):** Atrial and nodal premature beats
- **V (Ventricular):** Premature ventricular contractions
- **F (Fusion):** Fusion of normal and ventricular beats
- **Q (Unknown):** Paced beats and unclassifiable beats

Moody and Mark [12] analyzed the impact of the MIT-BIH database on the field, noting that it has been cited in over 7,000 publications and has enabled direct comparison of algorithms across research groups. However, they also cautioned about methodological pitfalls, particularly the importance of maintaining patient-independent evaluation to obtain realistic performance estimates. Our evaluation methodology directly addresses this concern through GroupKFold cross-validation.

Research Gap: Our work addresses the need for a computationally efficient pipeline that tightly integrates signal processing with ensemble classification, providing interpretable results without sacrificing accuracy.

3 Novel Aspects and Technological Contributions

Our project introduces the following innovations:

- **Integrated Stream-Based Methodology:** A unified pipeline where signal processing and ML are coupled via the “Anchor” concept (R-peak as coordinate origin for all features).
- **Hybrid Feature Set:** Combining physiological timing (RR intervals, QT duration), normalized amplitudes (P, Q, S, T waves), and PCA-based QRS morphology in a single feature vector optimized for tree-based classification.
- **Robust Denoising:** Use of db4 Wavelets with adaptive thresholding to preserve peak structures while removing baseline wander and muscle noise [2].
- **Reproducible Evaluation:** Patient-wise GroupKFold cross-validation ensuring realistic performance estimates, with documented configurations for reproducibility.

4 System Requirements

4.1 Functional Requirements

1. Ingest ECG data in WFDB/CSV formats
2. Apply band-pass filtering (0.5–40 Hz) and wavelet denoising (db4)
3. Detect R-peaks using Pan-Tompkins algorithm
4. Extract temporal features (RR intervals, QRS/QT duration)
5. Extract amplitude features (normalized P, Q, S, T voltages)
6. Extract morphological features (PCA coefficients)
7. Classify beats using Random Forest into 5 AAMI classes
8. Generate performance reports (confusion matrix, F1-scores)

4.2 Use Cases / User Stories

Use Case 1: Holter Analysis

- **Actor:** Clinical technician
- **Flow:** Load recording → Preprocess → Detect R-peaks → Extract features → Classify beats → Review flagged abnormalities

Use Case 2: Model Training

- **Actor:** Research engineer
- **Flow:** Specify dataset → Extract features → Train RF with GroupKFold → Report performance → Save model

4.3 Non-Functional Requirements

- **Accuracy:** $\geq 98\%$ overall; $\geq 90\%$ F1-Score for V class
- **Latency:** ≤ 100 ms per beat on CPU
- **Robustness:** $< 5\%$ accuracy drop at 10dB SNR

5 Project Plan

5.1 Resource Requirements

Hardware: Development laptops (≥ 8 GB RAM), Google Colab for experiments

Software: Python 3.9+, NumPy, SciPy, scikit-learn, PyWavelets, WFDB, imbalanced-learn

Data: MIT-BIH Arrhythmia Database (PhysioNet)

5.2 Work Breakdown and Work Assignment

Table 5.1: Work Package Assignment

Work Package	Main Tasks	Lead
WP1: Signal Processing	Wavelet denoising (db4), Pan-Tompkins implementation, R-peak correction	Mehmet Eren Şahin
WP2: Feature Extraction	PCA for morphology, interval logic (QT/ST), feature DataFrame construction	Melih Demir
WP3: Machine Learning	Random Forest implementation, hyperparameter tuning, SMOTE & GroupKFold validation	Mert Aydın

5.3 Time Plan

Table 5.2: Project Timeline

Period	Activities
Oct–Dec 2025 (Done)	Literature survey, MIT-BIH setup, Signal Processing (WP1)
Jan 2026 (Current)	Initial Random Forest training (WP3), Feature Importance analysis
Feb–Mar 2026	Advanced tuning, SMOTE integration, Patient-wise testing
Apr–May 2026	Robustness testing, Final Report

Gantt Chart								
Task	O	N	D	J	F	M	A	M
Literature Survey	•	•						
WP1: Signal Proc.		•	•					
WP2: Features			•	•				
WP3: ML				•	•			
Testing & Report						•	•	•

Figure 5.1: Project Gantt Diagram

6 Goals and Evaluation Criteria

6.1 Goals

1. Engineer an end-to-end pipeline mapping raw ECG to clinical labels
2. Demonstrate that Random Forest with hybrid features achieves high accuracy
3. Provide robust, patient-independent evaluation

6.2 Evaluation Criteria

1. Overall accuracy $\geq 98\%$ (patient-independent 5-fold CV)
2. V-class F1-Score $\geq 90\%$
3. S-class F1-Score $\geq 85\%$
4. Inference latency ≤ 100 ms per beat
5. Accuracy degradation $< 5\%$ at 10dB SNR noise injection

At the end of the project, even though performance will not be evaluated based solely on meeting these targets, we are expected to examine our system's performance against these criteria in the final report.

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